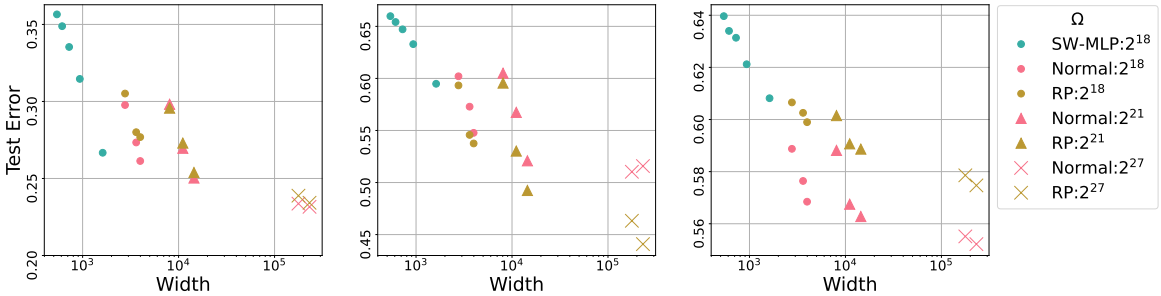

Figure 5. Test error improved as the effective width increased. This figure presents S-Mixer, RP S-Mixer, and SW-MLP models on CIFAR-10 (left), CIFAR-100 (center), and STL-10 (right). Experiments were conducted using three different random seeds, and the mean test error is depicted. The observed standard deviations were less than 0.026 for CIFAR-10, 0.056 for CIFAR-100, and 0.008 for STL-10.

MLP	CIFAR-10	CIFAR-100	max. width	#connections
β-LASSO (Neyshabur, 2020)	14.81	40.44	-	256M
Mixer-SS/8	15.91 ($\pm$1.55)	44.24($\pm$1.83)	6.3×10^4	256M
Mixer-SS-W	**12.07** ($\pm$0.47)	**38.13**($\pm$1.36)	$\mathbf{1.2 \times 10^5}$	255M

MLP	ImageNet-1k	max. width	Ω	S	C
Mixer-B/16 (Tolstikhin et al., 2021)	23.56	6.0×10^5	2.6×10^8	196	786
Mixer-B-W	**23.26** ($\pm$ 0.19)	$\mathbf{6.2 \times 10^5}$	2.6×10^8	256	588

Table 1. Test error on CIFAR-10/CIFAR-100/ImageNet-1k from scratch. (Upper Table) By setting S and C closer **under the same number of total connections** throughout layers, the maximal width of layers became larger in ours (Mixer-SS-W). Its test error eventually improved more than β-LASSO. (Lower Table) By setting S and C closer **under the same** Ω, the test error in ours (Mixer-B-W) improved than the original MLP-Mixer (Mixer-B/16). Each experiment is done with three random seeds.

5.2. Random Permuted Mixers

In normal Mixers, J_1 and J_2 are restricted to the identity or commutation. This means that the sparse weight matrices of the effective MLP are highly structured because their block matrices are diagonal. To destroy the structure, we introduce RP-Mixers. A *RP S-Mixer* has (J_1,J_2) in each PK layer, which is given by random permutation matrices as $U = \text{PK-Layer}_W(X; J_1, J_2)$ and $\text{PK-Layer}_{V^\top}(U; J_1', J_2')$. Similarly, for a *RP MLP-Mixer*, we set the PK layer corresponding to token mixing and channel mixing to the random permutation matrices. From the definition of the PK layer (5.1), this is equivalent to the effective MLP with width SC (and γSC for the MLP-Mixer) and sparse weight

$$W_{eff} = J_2(I_{n_1} \otimes W)J_1. \tag{19}$$

Because (J_1, J_2) are random permutations, the non-zero entries of W_{eff} are scattered throughout the matrix. In this sense, RP Mixers seemingly become much closer to random sparse weights than the normal Mixers. Figure 1(d) illustrates this scattered random weight configuration, while Figure S.9 presents an actual numerical example. Since the permutation matrices are orthogonal matrices, W_{eff}'s singular values remain identical to those of the normal Mixer, thereby preserving the spectrum discussed in Section 4.3.

5.3. Increasing width

Figure 5 shows that the test error improves as the effective width of the Mixers increases. We trained the normal and RP S-Mixers for various values of S and C with fixed Ω. The normal and RP S-Mixers show similar tendencies of increasing test error with respect to the effective width m. The normal and RP MLP-Mixers also show similar tendencies as is shown in Appendix D.2. Because the static-mask SW-MLP requires an $m \times m$ weight matrix, an SW-MLP with a large Ω cannot be shown in the figure. In contrast, we can use a large effective width for the Mixers, and the error continues to increase as the width increases. This can be interpreted as the Mixers realizing a width setting where naive MLP cannot reach sufficiently. Eventually, the figure suggests that such an extremely large width is one of the factors contributing to the success of the Mixer.

Table 1 shows a comparison of MLP-Mixer with a dynamic sparsity β-LASSO (Neyshabur, 2020). We found a wider Mixer (Mixer-SS-W) has better performance than β-LASSO. Table 1 shows a comparison of Mixer-B/16 (Tolstikhin et al., 2021) and a wider Mixer (Mixer-B-W). By setting S and C closer under fixed Ω, the maximal width of layers became larger in ours (Mixer-B-W). Its test error eventually improved more than the original MLP-Mixer (Mixer-B/16). In both results, the wideness improved the performance even if Ω is fixed.

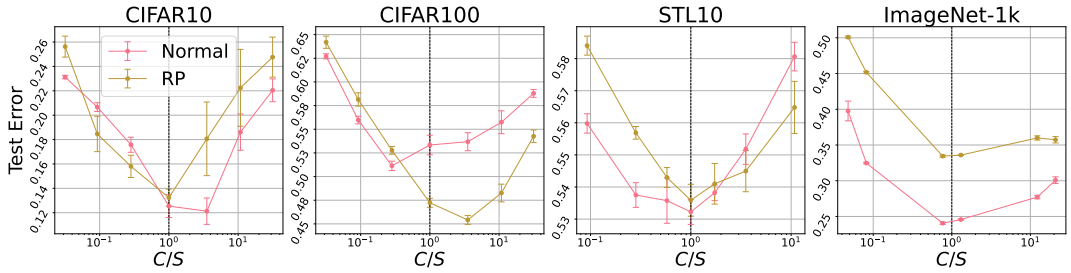

Figure 6. PK family achieves the lowest test error around $C = S$. S-Mixers on (a) CIFAR-10, (b) CIFAR-100, (c) STL-10. MLP-Mixers on (d) ImageNet-1k. Red line: Normal Mixers, yellow line: RP Mixers, dashed lines: $C = S$.

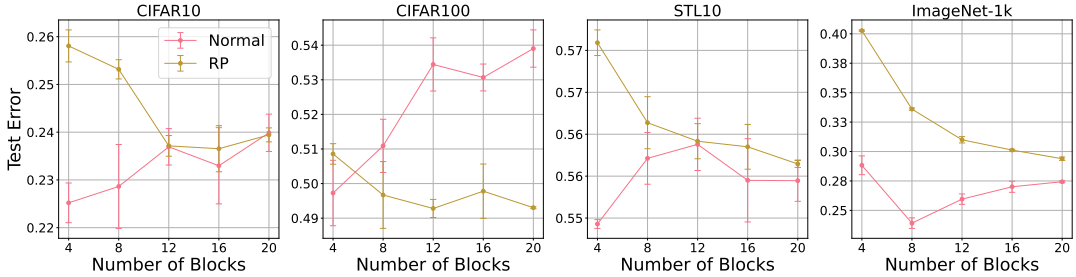

Figure 7. RP Mixers can become comparable to or even beat normal ones if the depth increases. We set $C = S = 128$.

5.4. Performance at the optimal width and sparsity

Figure 6 confirms that the maximum width (15) derived from Hypothesis 4.1 adequately explains the empirical performance of the Mixers. Models were trained using supervised classifications for each dataset. For CIFAR-10, CIFAR-100 and STL-10, we trained normal and RP S-Mixers. We fixed the dimension of the per-patch FC layer and changed S and C while maintaining a fixed number of Ω. It can be observed that the test error was minimized around $C = S$, as expected from Hypothesis 4.1 and (15). This tendency was common to the normal and RP Mixers. For ImageNet-1k, we trained normal and RP MLP-Mixers. Similarly, its performance is maximized around $C = S$. We observed a similar tendency also in different settings of hyparparamters in Appendix D.3 and Appendix D.5.

Remark on depth: As is shown in Figure 7, we observed that RP Mixers tended to underperform at limited depths but could achieve comparable or better results than normal Mixers with increased depths. This seems rational due to deep RP-Mixer's resistance to overfitting or shallow RP-Mixer's small receptive fields (see Appendix D.4 for the details). We also confirmed that this dependence on depth did not change the fundamental similarity regarding the width and sparsity.

6. Conclusion and future directions

This work provides novel insight that the MLP-Mixer effectively behaves as a wide MLP with sparse weights. The analysis in the linear activation case elucidates the implicit sparse regularization through the Kronecker-product expression and reveals a connection to Monarch matrices. The SW-MLP, normal and RP Mixers exhibit a quantitative similarity in performance trends, verifying that the sparsity is the key mechanics underlying the MLP-Mixer. Maximizing the effective width and sparsity leads to improved performance. We expect that the current work will serve as a foundation for exploring further sophisticated designs of MLP-based architectures and the efficient implementation of neural networks with desirable implicit biases.

Exploring the potential of MLPs with structured weights further will be interesting. Evaluating the optimal width and sparsity theoretically could be an exciting research topic (Edelman et al., 2023). As we noted, the solvability of global minima and dynamics in mixing layers, even with linear activation, remains uncertain, and the theory has yet to fully address or circumvent this issue. It would be also interesting to clarify whether other potential candidates for memory-friendly architectures with desirable inductive biases. In particular, weight sharing is known to perform well for image domains, occasionally outperforming networks without weight sharing (Ott et al., 2020). Given that the Mixers can be seen as approximations of MLPs with weight-shared Monarch matrices, it will be an interesting theme to evaluate the validity of such approximations.

References

Arora, S., Cohen, N., Hu, W., and Luo, Y. Implicit regularization in deep matrix factorization. *Advances in Neural Information Processing Systems*, 32, 2019.

Bachmann, G., Anagnostidis, S., and Hofmann, T. Scaling mlps: A tale of inductive bias. In *Advances in Neural Information Processing Systems*, 2023.

Bai, Z. and Silverstein, J. W. *Spectral analysis of large dimensional random matrices*, volume 20. Springer, 2010.

Bjorck, N., Gomes, C. P., Selman, B., and Weinberger, K. Q. Understanding batch normalization. *Advances in Neural Information Processing Systems*, 2018.

Cordonnier, J.-B., Loukas, A., and Jaggi, M. On the relationship between self-attention and convolutional layers. In *International Conference on Learning Representations*, 2019.

Cubuk, E. D., Zoph, B., Mane, D., Vasudevan, V., and Le, Q. V. Autoaugment: Learning augmentation strategies from data. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 113–123, 2019.

Dao, T., Chen, B., Sohoni, N. S., Desai, A., Poli, M., Grogan, J., Liu, A., Rao, A., Rudra, A., and Ré, C. Monarch: Expressive structured matrices for efficient and accurate training. In *International Conference on Machine Learning*, pp. 4690–4721. PMLR, 2022.

d'Ascoli, S., Sagun, L., Biroli, G., and Bruna, J. Finding the needle in the haystack with convolutions: on the benefits of architectural bias. *Advances in Neural Information Processing Systems*, 2019.

Dettmers, T. and Zettlemoyer, L. Sparse networks from scratch: Faster training without losing performance. *arXiv preprint arXiv:1907.04840*, 2019.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Edelman, B. L., Goel, S., Kakade, S., Malach, E., and Zhang, C. Pareto frontiers in neural feature learning: Data, compute, width, and luck. In *Advances in Neural Information Processing Systems*, 2023.

Evci, U., Gale, T., Menick, J., Castro, P. S., and Elsen, E. Rigging the lottery: Making all tickets winners. In *International Conference on Machine Learning*, pp. 2943–2952. PMLR, 2020.

Folland, G. B. *Real Analysis: Modern Techniques and Their Applications*. John Wiley & Sons, 2013.

Frankle, J. and Carbin, M. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*, 2019.

Fu, D. Y., Arora, S., Grogan, J., Johnson, I., Eyuboglu, S., Thomas, A. W., Spector, B. F., Poli, M., Rudra, A., and Re, C. Monarch mixer: A simple sub-quadratic gemm-based architecture. In *Advances in Neural Information Processing Systems*, 2023.

Gadhikar, A. H., Mukherjee, S., and Burkholz, R. Why random pruning is all we need to start sparse. In *International Conference on Machine Learning*, pp. 10542–10570. PMLR, 2023.

Golubeva, A., Neyshabur, B., and Gur-Ari, G. Are wider nets better given the same number of parameters? In *International Conference on Learning Representations*, 2021.

Hameed, M. G. A., Tahaei, M. S., Mosleh, A., and Nia, V. P. Convolutional neural network compression through generalized kronecker product decomposition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 36:771–779, 2022.

Hoff, P. D. Multilinear tensor regression for longitudinal relational data. *The annals of applied statistics*, 9(3): 1169, 2015.

Hoff, P. D. Lasso, fractional norm and structured sparse estimation using a hadamard product parametrization. *Computational Statistics & Data Analysis*, 115:186–198, 2017.

Hwang, J. Y., Lee, J. O., and Schnelli, K. Local law and Tracy–Widom limit for sparse sample covariance matrices. *The Annals of Applied Probability*, 29(5):3006 – 3036, 2019.

Liu, S., Chen, T., Chen, X., Shen, L., Mocanu, D. C., Wang, Z., and Pechenizkiy, M. The unreasonable effectiveness of random pruning: Return of the most naive baseline for sparse training. In *International Conference on Learning Representations*, 2022.

Magnus, J. R. and Neudecker, H. *Matrix differential calculus with applications in statistics and econometrics*. John Wiley & Sons, 2019.

Martens, J. and Grosse, R. Optimizing neural networks with Kronecker-factored approximate curvature. In *International Conference on Machine Learning*, pp. 2408–2417. PMLR, 2015.